

HTA Methodology & Process Summary

Clinical Efficacy vs. Macroeconomic System Access

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CASE STUDY PORTFOLIO

1 Project Overview

This technical brief outlines the strategic methodology, health economic parameters, and co-creation workflow used to construct the interactive Health Technology Assessment (HTA) case study for a first-in-class dual GIP/GLP-1 receptor agonist under **NICE Technology Appraisal [TA1026]** guidelines.

Developed by a clinician transitioning into strategic market access, this framework bridges the gap between individual patient outcomes and macro-level health system capacity. It demonstrates how complex medical trial literature is translated into scalable population-resource optimization models.

2 Development Methodology Flowchart

Below is the structured, end-to-end operational pathway followed to translate a static regulatory decision into a dynamic, interactive health technology assessment visualizer:



3 Step-by-Step Execution Breakdown

3.1 Phase 1: Clinical Evidence Discovery

- **Data Collection:** Appraised clinical endpoints from the landmark **SURMOUNT-1** trial (focusing on systemic GIP/GLP-1 metabolic regulation) and the **SURMOUNT-2** trial assessing patients with comorbid Type-2 Diabetes.
- **Clinical Synthesis:** Synthesized how a dual agonist behaves as a disease-modifying therapy, arresting long-term progression of macrovascular and microvascular co-morbidities.

3.2 Phase 2: Health Economic Parameter Mapping

- **The Cost-utility Standard:** Recognized that under standard parameters, an asset may be highly cost-effective relative to traditional lifestyle changes (within standard £20k–£30k/QALY margins).
- **The Prevalence Clash:** Because obesity affects approximately 1 in 4 adults in the UK, open reimbursement under standard marketing authorizations ($BMI \geq 30 \text{ kg/m}^2$) would capture roughly **3.4 million individuals**, resulting in an unsustainable, immediate national budget shock.

3.3 Phase 3: Mathematical Algorithm & Interactive Design

To model system-wide financial consequences under different access frameworks, a multi-tiered budget impact calculation was established:

1. **Target Population Base (P_{base}):** Configured a scalable model of 1.0M to 5.0M patients (standardizing a default baseline of 3.4M).
2. **Annual Drug Cost (C_{annual}):** Set a variable pricing range of £500 to £2,500 per patient.
3. **Cohort Restrictions (Eligibility Index E):**
 - *Unrestricted Pathway ($E = 1.0$):* Standard label indication ($BMI \geq 30$ with ≥ 1 comorbidity).
 - *NICE Restricted Guidance ($E = 0.35$):* Guarded clinical access (severe obesity with $BMI \geq 35$ and ≥ 1 comorbidity).
 - *NHS Year 1 Phase ($E = 0.05$):* High-risk priority rollout ($BMI \geq 40$ with multiple comorbidities).

System Budget Impact Formulas:

The Gross Annual System Budget Impact (I_{gross}) is calculated as:

$$I_{\text{gross}} = P_{\text{base}} \times E \times C_{\text{annual}} \quad (1)$$

The Resource Savings/Opportunity Cost Avoided (S_{avoided}) relative to the unrestricted market path is defined as:

$$S_{\text{avoided}} = (P_{\text{base}} \times C_{\text{annual}}) - I_{\text{gross}} \quad (2)$$

These formulas demonstrate how restricting access to severe cohorts aligns the overall budget impact with public health system funding capacities.

3.4 Phase 4: Generative AI Co-Creation & UI Build

* **Rapid Prototyping:** Leveraged Generative AI as an architectural partner, using structured code blocks to assemble a single, high-fidelity responsive HTML document with no external dependencies.

* **Interactive Visual Engine:** Co-designed a custom, dual-column interactive container using modern CSS flexbox grids and clean vanilla JavaScript events, providing instantaneous mathematical visual rendering of changing economic variables.

Why did I create this interactive HTA case study?

By combining this interactive health economic case study with my ongoing preparation for advanced credentials in Health Economics, I wanted to challenge myself to look beyond individual clinical outcomes and understand the broader picture of macro-level health system value.

Building this project was a hands-on way to practice translating complex clinical evidence into structured, population-level resource optimization models. As a clinician transitioning into the strategic market access space, I am deeply curious about how health technologies are evaluated at scale. This asset represents my first steps in bridging the gap between frontline patient care and healthcare economics.